Where Internationalization Pays: Institutional Regime and Digital Capability as Joint Determinants of the Performance Optimum across 45 Asian and Pacific Economies

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Abstract

The central unresolved question in research on the internationalization–performance (I–P) relationship is not whether it is non-linear, but what determines where its performance-maximising optimum falls. We develop and test an integrated multi-level account in which the inverted-U is universal in form but jointly conditioned by a firm's institutional regime and digital capability, the regime governing the curve's amplitude and digital capability reshaping it as a partial substitute for institutional strength. Using harmonised World Bank Enterprise Survey microdata (baseline N = 82,302; 45 economies; 98 country-year waves, 2003–2025) spanning all six Institutional Context Regime Variation (ICRV) groups, we estimate hierarchical OLS models. The I–P relationship is robustly inverted-U with a turning point near 37% export intensity (Lind–Mehlum $p < .001$). Institutional regime moderates this optimum: concavity deepens as institutional quality declines, so the over-internationalisation penalty is sharpest in frontier and vulnerable regimes. Digital adoption both raises performance and reshapes the curve, its reshaping potency significantly stronger in weaker-institution regimes (DAI × ICRV $p = .049$), consistent with digital-for-institutional substitution. The account resolves the fragmented Asian I–P literature: divergent country findings locate firms at different positions on a single optimum jointly set by institutional quality and digital capability.

Keywords: internationalization–performance; inverted-U; institutional regime; digital capability; Asia-Pacific; World Bank Enterprise Surveys

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1. Introduction

The same export-intensity decision that lifts a firm in one economy depresses it in another: as a firm moves along the institutional gradient, the performance payoff to internationalization slides predictably. This *institutional slope*, not the sign of the average effect, is the organizing puzzle of this paper. A central question in international business research concerns whether and how export internationalization translates into firm performance (Lu & Beamish, 2004; Johanson & Vahlne, 2009). Meta-analyses document a mean correlation of approximately r = .07 across several thousand studies (Bausch & Krist, 2007; Kirca et al., 2012; Marano et al., 2016), yet the variance across studies vastly exceeds what sampling error can explain. In Asia, a region that accounts for over half of global manufacturing exports (UNCTAD, 2023), the evidence is especially fragmented: prior studies report positive relationships (Pangarkar, 2008; Korea), inverted-U curves (Chiao et al., 2006; Taiwan), or null and negative effects (Mondal et al., 2022; India). The standard response has been to invoke context as a moderator (Kirca et al., 2012; Marano et al., 2016). Yet treating context as a moderator leaves the deeper question unanswered: if the I–P relationship is non-linear, what determines *where* its performance-maximising turning point falls, and why should that location differ systematically across firms and economies? Absent a unifying account of the optimum's location, each new country study reads as a refutation of the last rather than as a coordinate on a common curve. Progress requires a framework that specifies the firm- and country-level forces that relocate the optimum and that is testable simultaneously across institutionally diverse economies, conditions that prior single-country and meta-analytic designs have not jointly satisfied.

This paper addresses that gap. We pool 98,658 firm-observations from 45 Asian and Pacific economies spanning 2003–2025, harmonised from the World Bank Enterprise Survey (WBES); after imposing non-missing productivity and export-intensity requirements, the baseline estimation sample is 82,302 firm-observations (falling to 28,500 in the fully specified model M11). Our sample spans all six ICRV regime groups, from advanced innovation-driven economies (Singapore, Korea, Taiwan, HongKong, Israel) to Pacific Small Island Developing States (Vanuatu, Solomon Islands, Fiji, Tonga, Samoa, Kiribati, Papua New Guinea, Timor-Leste) and Gulf advanced-resource economies (Bahrain, Kuwait, Qatar, Brunei, Saudi Arabia), classified through the Institutional Context Regime Variation (ICRV) framework that anchors a broader multi-country research programme of which this study is part.

Our contributions are theoretical, not merely empirical. First, we advance a *relocated-optimum* account of the I–P relationship: the inverted-U is universal in form, but its curvature is endogenous to the institutional regime, deepening as institutional quality declines (confirmed across specifications at N = 82,302–28,500; M11: TP = 25.7%, Lind–Mehlum p = .026). This reframes the long-running "is it an inverted-U?" debate as a question about the *determinants of the optimum's location*, and reconciles otherwise conflicting country evidence within a single curve. Second, we theorise and test *digital-for-institutional substitution*: the curve-reshaping potency of digital adoption is significantly greater where formal institutions are weaker (DAI $\times$ ICRV, p = .049), positioning foundational digital capability as a partial substitute for institutional infrastructure in converting exports into performance, a multi-economy extension of the institutional-voids logic (Khanna & Palepu, 1997) into the digital domain. Third, we formalise the Institutional Context Regime Variation (ICRV) framework as a firm-relevant contingency construct that orders economies jointly by institutional capability and exposure to resource vulnerability, and show that it predicts both the slope and the curvature of the I–P function, distinguishing the curve-*relocating* role of institutions (H5) from the level-*shifting* role of firm capabilities such as technological capability (TCI). Taken together, these contributions move the multi-country I–P literature from cataloguing heterogeneous effects toward a parsimonious, jointly-determined account of when and where internationalization pays.

The remainder of the paper is organised as follows. Section 2 develops the theoretical framework and formal hypotheses. Section 3 describes data and methods. Section 4 presents results. Section 5 discusses implications and limitations. Section 6 concludes.

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2. Theoretical Framework and Hypotheses

2.1 Theoretical Foundations

We draw on four theoretical traditions. The Uppsala model (Johanson & Vahlne, 1977, 2009) describes internationalization as a sequential process in which firms accumulate experiential knowledge, progressively reducing psychic distance and coordination costs. The Resource-Based View (Barney, 1991; Wernerfelt, 1984) positions firm-level capabilities, technological, digital, and managerial, as the mediating mechanism linking export exposure to performance outcomes. Institutional Theory (North, 1990; Scott, 2008) directs attention to how country-level formal and informal institutions shape the cost structure of cross-border transactions. Upper Echelons Theory (Hambrick & Mason, 1984; Hambrick, 2007) emphasises that top management team attributes moderate how firms respond to strategic opportunities and constraints. Rather than four parallel traditions, this paper integrates them through one organizing spine: the ICRV institutional context functions as a *transaction cost filter* that defines a *strategic export topography* on which the I–P optimum sits. Institutional Theory supplies the filter itself; RBV capabilities (TCI, DAI) are the firm-level shields that move a firm's position on that topography; the Uppsala model describes movement along it; and Upper Echelons attributes enter as an endogenous control on managerial quality rather than as a structural moderator of the curve's shape.

These traditions are typically invoked piecemeal. We integrate them through a single contingency construct, the **Institutional Context Regime Variation (ICRV)** framework, that specifies *where* on the I–P curve a firm's environment places it. ICRV orders economies along two theoretically distinct dimensions that prior typologies treat separately: *institutional capability* (the formal-institutional quality emphasised by North (1990) and elaborated in the varieties-of-capitalism tradition; Hall & Soskice, 2001) and *resource vulnerability* (exposure to external shocks, narrow factor endowments, and the institutional voids that firms must internalise; Khanna & Palepu, 1997). The resulting six regime groups range from advanced innovation-driven economies, through resource-driven, upper-middle, emerging, and frontier economies, to highly vulnerable small-island states. Our central theoretical claim follows from locating the firm within this regime space: the inverted-U *form* of the I–P relationship is universal (H1), but its shape and performance-maximising optimum are *jointly conditioned* by the institutional regime (H5) and by the firm's digital capability (H3), with digital capability acting as a partial substitute for institutional strength where the latter is weak (H6). Firm capabilities enter on a second axis: technological capability is hypothesised to *amplify* the curve (H2) and managerial characteristics to lift the performance *level* without reshaping it (H4); whether each in fact relocates the optimum or merely shifts the level is an empirical question the design is built to adjudicate. This contrast between optimum-*relocating* forces (institutional regime and digital capability) and firm-*capability* forces (technological and managerial) is the analytical spine of the paper.

2.2 The Non-Linear I–P Relationship (H1)

Contractor et al. (2003) formalised the three-stage S-curve hypothesis: early internationalization carries fixed entry costs that depress performance; intermediate internationalization yields learning and scale economies; high internationalization may again impose coordination and agency costs. Empirically, however, the data more consistently support the inverted-U, a special case in which the third stage does not emerge at the export intensities typically observed in developing-economy WBES data (Gomes & Ramaswamy, 1999; Lu & Beamish, 2004).

In the Asian context, three companion country-specific studies provide prior evidence. For Vietnam (companion study), the inverted-U is confirmed with a turning point at 39–46% using a WBES-based instrumental variable design. For China (companion study), an inverted-U turning point of 47–49% is replicated across specifications with Paternoster cross-cohort tests ($p = .545$). For Singapore (companion study), the relationship is predominantly positive with a very late turning point (~82%), consistent with near-monotonic returns in a high-institutional, digitally saturated economy. Pooled across a much larger and institutionally heterogeneous sample, we expect the inverted-U to emerge at an intermediate turning point.

> **H1:** The relationship between export intensity (FSTS) and firm labour productivity (ln LP) is an inverted-U, with a statistically confirmed turning point at intermediate export intensities, in a pooled sample of Asian and Pacific firms.

2.3 Technological Capability as a Curve Amplifier (H2)

Technological capability, operationalised as a composite of quality certification, foreign-technology adoption, process innovation, and R&D, raises absorptive capacity (Cohen & Levinthal, 1990), enabling firms to extract greater performance value per unit of export exposure. Critically, we expect TCI to operate as a *curve amplifier*: at low export intensities, high-TCI firms gain more per unit of internationalization (steeper ascending limb); at high intensities, they experience sharper diminishing returns as coordination demands exceed even their superior capabilities (steeper descending limb). The net effect is a shift and steepening of the inverted-U, not just a parallel upward shift.

This amplification logic carries a built-in scope condition that the multi-economy design lets us test directly. Curve reshaping, as opposed to a parallel level lift, requires that the marginal productivity return to technological capability itself vary along the export-intensity range, which in turn presumes an institutional environment that lets capability translate into differential cross-border returns: enforceable contracts, protectable intellectual property, and functioning technology-transfer channels. Where those complements are present, the amplifier mechanism should operate; where they are unevenly distributed, as they necessarily are across forty-five economies spanning six institutional regimes, firm-level capability and the curvature it would reshape are no longer aligned, and the amplification predicted within any single institutional setting may average out in the pool. We therefore advance H2 as a curve-amplifier hypothesis while recognising a competing prediction, that in an institutionally heterogeneous pool TCI survives as a robust level shifter but its curvature-reshaping signal attenuates. Adjudicating between these is itself informative: a null curvature interaction in the pool, set against the curve moderation documented in single-country studies, would localise capability-driven reshaping as an institutionally contingent rather than universal phenomenon.

This prediction is consistent with the companion Vietnam study (β(TCI) = +0.179, IV-identified) and the companion China study (β(TCI) increasing from +0.28 to +0.43 across cohorts), both showing TCI as a positive level shifter. the companion Singapore study shows negligible TCI moderation of the curve, likely because near-universal adoption in a high-ICRV economy leaves insufficient variance for moderation.

> **H2:** TCI amplifies the I–P curve, raising the average performance level and steepening both the ascending and descending limbs of the inverted-U.

2.4 Digital Adoption as a Level Enhancer (H3)

Digital adoption, measured through website presence and electronic-payment share, has been associated with reduced communication and transaction costs (Verhoef et al., 2021). In the CDCM (Context-Contingent Digital Capability Model) developed in related work in this programme, DAI's effect depends on the interaction of regime quality, FSTS level, and digital market saturation (Authors, 2024). Specifically, in institutionally advanced economies where digital infrastructure is mature, DAI provides universal performance lifts (as in the companion Singapore study: β(DAI) positive and significant). In emerging-economy contexts with substantial selection into digital adoption (the companion Vietnam study: IV-estimated $\beta(\mathrm{DAI}) \approx 0$, consistent with selection bias in OLS), DAI's curve-shaping role is attenuated. Across the pooled multi-economy sample, we expect DAI to raise performance levels broadly across the sample (level effect). In a fully specified model that also accounts for managerial quality, which correlates with digital adoption, the residual DAI variation may reveal a capability-complementarity curve effect: digitally equipped firms at low FSTS face sunk adoption costs that compress near-term productivity, while digital infrastructure supports coordination efficiency at high FSTS where cross-border management demands are greatest.

The predicted curve signature, a negative FSTS $\times$ DAI interaction paired with a positive $\mathrm{FSTS}^2 \times \mathrm{DAI}$ interaction, admits two non-exclusive readings that the present design cannot fully separate, and we flag the ambiguity rather than assume it away. The first is a substantive coordination mechanism: foundational digital interfaces lower the marginal cost of managing geographically dispersed transactions, so their performance contribution is small for domestically oriented firms but rises precisely where cross-border coordination demands are heaviest, flattening the descending limb. The second is a selection or sunk-cost artefact: firms that adopt low-cost digital tools may be disproportionately those facing binding productivity constraints, or may carry near-term adoption costs not yet amortised, depressing the ascending-limb slope without any causal coordination benefit. Both mechanisms generate the same observable curvature, so a significant interaction is consistent with either; we return to this identification limit in the discussion and treat the level effect, for which selection works against rather than toward the hypothesis, as the more robustly interpretable component.

> **H3:** DAI has a positive main effect on labour productivity and, when managerial characteristics are controlled, reshapes the I–P curve by compressing the ascending limb at low export intensities and attenuating performance decline at high export intensities.

2.5 Managerial Characteristics (H4)

Upper Echelons Theory predicts that top manager experience and gender composition affect strategic cognition and decision-making (Hambrick & Mason, 1984). Manager experience reduces coordination errors in cross-border operations; female top management is associated with stakeholder orientation, relationship capital, and risk management, all relevant in institutionally heterogeneous export markets (Richard et al., 2019). We expect both to raise the performance level for any given FSTS. However, given the breadth of the multi-country sample, firm-level managerial differences are unlikely to systematically reshape the aggregate FSTS–performance curve, whose shape is primarily determined by economy-wide institutional and market-maturity factors.

> **H4:** Top manager experience and female top manager status have positive main effects on firm performance, but do not significantly moderate the FSTS–performance curve shape.

2.6 Institutional Regime as a Turning-Point Shifter (H5)

North (1990) established that formal institutions reduce transaction costs; stronger institutions lower the cost of cross-border coordination and thus shift the performance-maximising export intensity (the inverted-U turning point). In higher-quality institutional environments, firms need a lower export share to recoup internationalization costs, so the turning point occurs earlier. Equivalently, holding FSTS constant, firms in stronger institutional regimes earn higher returns per unit of internationalization.

We operationalise institutional regime through the ICRV classification system developed in related work in this programme: six groups ranging from Group I (Advanced innovation-driven: Singapore, Korea, Taiwan, Israel) to Group VI (SIDS and small open economies: Vanuatu, Solomon Islands, Timor-Leste). Higher ICRV group numbers denote weaker institutions.

It is important to separate two observationally distinct ways in which institutions can moderate an inverted-U, because they carry different empirical signatures and the prior literature has not kept them apart. The first is *location* moderation: institutions shift the performance-maximising turning point along the FSTS axis, so the optimum occurs at a systematically different export share across regimes. The second is *amplitude* moderation: institutions govern the concavity, or sharpness, of the curve, holding the approximate location of the peak roughly constant while making the over-internationalisation penalty steeper or flatter. A transaction-cost reading of North (1990) is consistent with either: weaker institutions raise the coordination and contracting costs that bend the curve downward at high export commitment, which can manifest as an earlier peak, a sharper concave decline, or both. Because the two signatures are not equivalent, we state H5 at the level of curve moderation and treat the *form* the moderation takes, location versus amplitude, as an empirical question the multi-economy design is built to adjudicate. We expect positive FSTS $\times$ ICRV and negative FSTS^2 $\times$ ICRV interactions in Group III–VI relative to Group I (a higher Group I baseline coefficient), the negative quadratic interaction implying a deepening concavity as institutions weaken.

> **H5:** The ICRV institutional regime systematically moderates the I–P curve. As institutional quality declines (higher ICRV group number), the concavity of the inverted-U deepens, so the marginal penalty to over-internationalisation is sharper in weaker-institution regimes; whether this operates primarily by relocating the turning point or by amplifying the curvature is adjudicated empirically.

2.7 Regime-Contingent Digital Effects (H6)

The Context-Contingent Digital Capability Model (CDCM) developed in this research programme predicts that the I–P curve-reshaping role of digital adoption is not uniform across institutional environments. In institutionally advanced economies (ICRV Group I–II), digital infrastructure is near-universal and close to the competitive baseline, the marginal performance differentiation from website and e-payment adoption is accordingly lower. In institutionally weaker economies (ICRV Group IV–VI), digital adoption remains differentiating: digital market access tools lower the liability-of-foreignness cost and provide cross-border coordination support that institutional voids cannot substitute. The DAI $\times$ ICRV interaction therefore tests whether digital capabilities' curve-reshaping potency (established in H3) varies systematically across the institutional spectrum, specifically, whether compressing the ascending limb and buffering performance decline at high FSTS are more pronounced in weaker-institution economies.

This prediction is consistent with the digital-complementarity strand of institutional theory (North, 1990; Stallkamp & Schotter, 2021): where formal institutions provide reliable contract enforcement, IP protection, and market information, digital infrastructure adds incrementally; where institutional voids prevail, digital tools provide a non-trivial substitute for missing formal mechanisms. H6 therefore does not predict stronger *level* effects in weak-institution contexts (which could reflect many confounds), but specifically predicts a stronger *curve-shaping* interaction between digital adoption and export intensity, a conditional complementarity claim.

> **H6:** The ICRV institutional regime moderates the digital-adoption curve-reshaping effect: digital adoption's compressive effect on the I–P ascending limb and its buffering effect at high FSTS (H3) are stronger in weaker institutional environments (higher ICRV group number), where digital infrastructure provides greater competitive differentiation in cross-border market access.

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3. Data and Methods

3.1 Data Source

Data derive from the World Bank Enterprise Surveys (WBES), a stratified random-sample survey of formal private-sector enterprises conducted periodically across over 130 countries. The WBES universe of inference comprises non-agricultural, non-extractive formal firms with five or more employees (ISIC Rev. 3.1/4 manufacturing, construction, retail, and other services), stratified by sector, firm size (small 5–19, medium 20–99, large 100+ employees), and sub-national location; sample sizes are set to achieve 7.5% precision at 90% confidence (a minimum of 120 firms per stratum), and design weights are applied for population-representative estimates (World Bank, 2022). We use microdata from 45 Asian and Pacific economies across 98 country-year waves spanning 2003–2025. This coverage encompasses six UN M.49 Asian sub-regions: East Asia (China, HongKong, Korea, Mongolia, Taiwan), Southeast Asia (Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Vietnam), South Asia (Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka), Central Asia (Kazakhstan, Kyrgyz Republic, Tajikistan, Turkmenistan, Uzbekistan), West Asia (Bahrain, Iraq, Israel, Jordan, Kuwait, Lebanon, Qatar, Saudi Arabia, Yemen), and Pacific/SIDS (Fiji, Kiribati, Papua New Guinea, Samoa, Solomon Islands, Tonga, Vanuatu).

The final analytical dataset contains 98,658 firm-observations (before applying outcome and FSTS non-missing requirements); imposing those requirements yields the M2 baseline estimation sample of 82,302 firm-observations, falling to 28,500 in M11 as digital-capability and managerial controls are added. We adopt a geographical definition of Asia following the United Nations Statistics Division M49 standard (United Nations Statistics Division, 2024), under which Asia comprises the Eastern, South-eastern, Southern, Central, and Western Asian sub-regions; the Pacific is added because the study frame is explicitly Asia-Pacific. To keep this geographical frame internally consistent, we exclude economies that straddle the Europe–Asia boundary and are conventionally treated as transcontinental: Türkiye and Cyprus, and the three South Caucasus economies (Armenia, Azerbaijan, Georgia), all of which the World Bank operationally files under Europe & Central Asia rather than Asia (World Bank, 2025). Comoros (Sub-Saharan Africa) is likewise excluded. Central Asia and the West-Asian Gulf and Levant are retained as undisputed parts of the Asian landmass. (A strict UN-M49 specification that re-includes Türkiye and Azerbaijan was examined and fails to identify the inverted-U, reinforcing the transcontinental exclusion.) Oman's single available vintage (2003) lacks the productivity, export-intensity and digital-capability items and therefore does not enter the estimation sample. WBES designs are cross-sectional within each wave, with standardised instruments ensuring common variable definitions across economies and years. We apply three harmonisation steps: (1) variable-priority mapping for candidate items across PICS3 (2009–2013), Standardised (2014–2018), and BREADY/BEE (2019–2025) questionnaire generations; (2) encoding-robust reading (primary UTF-8 with Latin-1/CP1252 fallback); (3) deduplication, selecting the largest file per (country, year) pair when multiple uploads exist.

3.2 Variables

Dependent variable. Labour productivity is operationalised as log annual sales per permanent worker: ln(LP) = ln(sales / l1_workers). Sales are taken from WBES item n3 (or d2 where n3 is missing). Currency effects are absorbed by country-year fixed effects in M5; without FE, productivity levels vary in nominal terms across economies and years.

Foreign sales intensity (FSTS). Defined as the share of sales from direct exports (d3c / 100), ranging from 0 to 1. We mean-centre FSTS at the sample mean for quadratic and interaction models ($FSTS_c$); the quadratic term $FSTS_c^2$ is computed from the centred value.

Technological capability index (TCI_z). A two-item composite (available in >95% of files) of quality certification (b8: has ISO or other certification) and foreign-technology adoption (e6: uses foreign-licensed technology), standardised to mean 0, SD 1. The z-score preserves cross-economy variation.

Digital adoption index (DAI_z). Website presence (c22b / e1) averaged with e-payment share (k33 / 100 where available), standardised to mean 0, SD 1. In waves without e-payment items, DAI equals website presence only.

Manager characteristics. Top manager experience in sector (b7, in years) and a binary indicator for female top manager (b7a / b6a).

Controls. Female ownership share (b4 > 0.5: majority female-owned, binary), foreign ownership percentage (b6a / 100), firm age (in years, current year − b5), and log permanent employees (ln l1_workers) as a size proxy.

Institutional regime (ICRV group). Integer 1–6 based on the ICRV classification; used as a continuous moderator in M10/M11 and as a group-level moderator for marginal effects plots.

3.3 Estimation Strategy

We estimate a hierarchical sequence of 11 models (Table 2). All models use OLS with HC1 heteroskedasticity-robust standard errors (Long & Ervin, 2000), clustering by country-year. Country-year fixed effects (M5) provide the most conservative benchmark. We prefer HC1 over clustered SE given that WBES sampling is stratified within, not across, country-years.

The inverted-U shape is formally confirmed using the Lind–Mehlum (2010) test, which requires: (a) $\beta_1 > 0$, $\beta_2 < 0$, and (b) the turning point lies strictly within the observed FSTS range, and (c) the slope at the extremes satisfies the directional prediction. We report the turning point as: TP = $-\beta_1 / (2\beta_2)$ + mean(FSTS), where β_1 and β_2 are estimated on the mean-centred $FSTS_c$ scale and the result is expressed as a proportion (0–1) of total sales.

For moderation tests (M7–M11), we include interaction terms FSTS$\times$moderator and FSTS$^2\times$moderator. The joint significance of the two interaction terms is tested with an F-test; we report both the joint p-value and individual coefficient p-values.

Identification and the interpretation of estimates. The WBES is cross-sectional within each wave, so our coefficients describe within-survey-year associations rather than panel-tracked causal effects, and we adopt the disciplined-language posture recommended for design-constrained inference (Meyer, van Witteloostuijn, & Beugelsdijk, 2017), reporting associations throughout. Two endogeneity channels warrant explicit attention. The first is selection into exporting: more productive firms are more likely to export and to export intensively, which biases the *linear* FSTS coefficient upward. This channel, however, operates primarily on the level and slope rather than on the sign of the curvature; for selection to manufacture a spurious inverted-U it would have to induce a non-monotone relationship between unobserved productivity and export intensity, a far more demanding condition than simple positive sorting. The second is reverse causality between digital adoption and productivity, which would inflate the DAI *main* effect; because selection of productivity-constrained firms into low-cost digital tools (Section 2.4) works *against* a positive level coefficient, the estimated DAI premium is, if anything, conservative. We bound these concerns three ways: the country-year fixed-effects benchmark (M5) absorbs all unobserved time-invariant and wave-specific country heterogeneity, including price levels, exchange-rate regimes, and macro shocks, and the inverted-U survives it (TP = 39.5%, LM $p < .001$); the firm-level instrumental-variable design deployed in the Vietnam study (companion study) returns a null instrumented DAI coefficient, corroborating that the OLS digital premium is partly selection-driven; and the curvature result is reproduced under the most demanding three-way specification (M11). A firm-level IV is not available at scale across all forty-five economies, which we record as a boundary on causal interpretation.

3.4 Sample Sizes

The analytic sample for M2 (outcome + FSTS non-missing) is N = 82,302. Adding controls reduces the sample to N = 36,772 (M3); country-year FE model M5 uses the same sample. TCI-moderation models (M6, M7) use $N \approx 36{,}481$; adding DAI (M8) $N \approx 36{,}372$; manager models (M9) $N \approx 34{,}228$; ICRV-moderation (M10) $N \approx 30{,}360$; full three-way (M11) $N \approx 28{,}500$. The drop from M2 to M3/M5 reflects missingness in control variables (primarily foreign ownership and firm age), which is more prevalent in smaller economies and early WBES waves.

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4. Results

4.1 Descriptive Statistics and Correlations

The full analytic sample spans 45 economies and 98 country-year waves. The mean FSTS across firms is approximately 10% (median $\approx$ 0), reflecting the fact that most WBES firms are domestically oriented; exporters (FSTS > 0) represent approximately 18% of the sample. Labour productivity ranges from ln(LP) $\approx$ 9 (Kiribati, Timor-Leste) to ln(LP) $\approx$ 20 (Vietnam, Korea, reflecting PPP-unadjusted nominal sales differences). Pairwise correlations among focal variables are modest (|r| < .15 for all pairs), and variance inflation factors in the full model are below 3.5, ruling out multicollinearity concerns.

4.2 Main Effect: Inverted-U (H1)

Table 2, Models M0–M5.

The baseline linear model (M0) confirms that FSTS alone adds no explanatory power (adjR^2 = .000). Adding the quadratic term FSTS^2 (M2) yields β_1 = +1.211 (p < .001) and β_2 = −1.630 (p < .001), consistent with an inverted-U. Following Haans, Pieters, and He (2016), all four formal conditions for a genuine inverted-U are satisfied in M2: (C1) β_1 = +1.211 (p < .001), confirming a positive ascending limb; (C2) β_2 = −1.630 (p < .001), confirming concavity; (C3) the turning point TP* $\approx$ 37.1% lies well within the observed FSTS range [0%, 100%]; and (C4) the marginal effect of internationalization is positive at the lower data boundary (FSTS $\approx$ 0%, slope $\approx$ +1.21) and negative at the upper boundary (FSTS $\approx$ 100%, slope $\approx$ −2.05), with opposite signs confirmed by the Lind–Mehlum (2010) u-test (joint p < .001). The inverted-U remains significant when controls are added (M3: TP = 34.1%, LM p < .001) and when country-year fixed effects absorb all unobserved time-varying country heterogeneity (M5: TP = 39.5%, LM p < .001, adjR^2 = .671). The consistency of the turning point across M2, M3, and M5, ranging from 34.1% to 39.5%, provides strong evidence that the inverted-U is not an artefact of omitted country-level confounders. The inverted-U is also confirmed in the full three-way moderation model M11 (TP = 25.7%, LM p = .026), establishing robustness to the most demanding specification.

H1 is supported. The I–P relationship is robustly inverted-U in the pooled sample of 45 Asian and Pacific economies, with a turning point of approximately 37% foreign sales intensity.

4.3 Technological Capability Moderation (H2)

Table 2, Models M6–M7.

Adding TCI as a main effect (M6: β = +0.342, p < .001) raises adjR^2 from .017 to .027. The interaction model M7 adds FSTS × TCI and FSTS^2 × TCI. Results show:

- TCI main effect: β = +0.372 (p < .001), higher TCI raises baseline performance
- FSTS × TCI: β = −0.090 (NS), the ascending limb interaction is not significant
- FSTS^2 × TCI: β = −0.227 (NS in M7; remains NS in M8, p = .077)

The joint F-test for the two TCI interaction terms is not significant across M7–M9. The inverted-U remains confirmed in M7 (TP = 34.2%, LM p < .001). A one-SD increase in TCI raises labour productivity by exp(0.372) − 1 ≈ 45% at mean FSTS, a large and practically meaningful level effect. The curvature steepening predicted by H2 does not emerge in this expanded sample.

H2 is partially supported (level effect confirmed; curvature effects NS). TCI raises the performance level consistently across specifications, but neither the ascending nor descending limb amplification reaches significance. The null curvature moderation likely reflects heterogeneous TCI effects across the institutionally diverse 45-economy sample, attenuating individual-country patterns documented in the companion Vietnam and China studies.

4.4 Digital Adoption Moderation (H3)

Table 2, Model M8.

When DAI is added to M7, the DAI main effect is positive and statistically significant (β = +0.172, p < .001), consistent with approximately a 19% performance premium for digitally active firms (exp(0.172) − 1 ≈ 19%). Strikingly, both DAI interaction terms are statistically significant in M8: FSTS × DAI = −0.588 (p < .001, ***) and FSTS^2 × DAI = +0.726 (p = .002, ***)*. The interaction pattern, negative FSTS × DAI and positive FSTS^2 × DAI, means that high-DAI firms experience a flatter, shifted I–P curve: lower productivity gains per unit of FSTS at low export intensities (sunk adoption costs, or selection of productivity-constrained firms into digital adoption), but significantly smaller performance declines at high FSTS (digital infrastructure supporting cross-border coordination at scale). This result strengthens further in M9 with manager controls: FSTS × DAI = −0.786 (p < .001,* **) and FSTS^2 × DAI = +0.982 (p < .001,** *). The inverted-U is maintained in both specifications (M8: TP = 34.2%; M9: TP = 32.2%).

H3 is supported. DAI raises firm performance by approximately 19% and significantly reshapes the I–P curve, compressing the ascending limb and attenuating the performance decline at high FSTS, even before controlling for managerial characteristics. This is the primary capability-moderating finding of the study.

4.5 Managerial Characteristics (H4)

Table 2, Model M9.

Manager experience (β = +0.018, p < .001) and female top manager (β = +0.202, p < .001) both raise firm performance independently of FSTS in M9. The experience effect implies approximately 2% higher labour productivity per additional year of sectoral experience; the female top manager effect implies approximately a 22% performance premium ($\exp(0.202) - 1 \approx 22\%$). The FSTS$\times$manager_experience interaction is not significant in either M9 (β = +0.005, p = .29) or M11 (β = +0.007, p = .26), indicating that managerial experience operates as a level shifter rather than a moderator of the I–P curve.

H4 is partially supported. Top manager experience and female management improve firm performance (level effects confirmed). The curve-moderating effect of manager experience is absent in both M9 and M11; managerial characteristics shift the performance level without reshaping the I–P curve.

4.6 Institutional Regime Moderation (H5)

Table 2, Model M10.

The ICRV group main effect (β = +0.592, p < .001) indicates that, controlling for TCI, DAI, and firm characteristics, each step up the ICRV scale (from Group I to VI, i.e., from stronger to weaker institutions) is associated with approximately an 81% higher labour productivity level ($\exp(0.592) - 1 \approx 81\%$). This counter-intuitive *positive* ICRV coefficient reflects that nominal sales-per-worker is higher in economies with lower price levels (typical of frontier/emerging economies), and country-year fixed effects were not included in M10 to preserve the institutional regime as the key moderator of interest.

More relevant for H5 are the interaction terms: FSTS $\times$ ICRV (β = +1.849, p < .001) and FSTS^2 $\times$ ICRV (β = −2.932, p < .001). The negative FSTS^2 $\times$ ICRV term indicates that the *concavity* of the I–P curve deepens as institutions weaken (higher ICRV group): the inverted-U is sharpest in frontier and vulnerable regimes and flattest, even mildly convex within the observed range, in the strongest-institution group. Decomposing by ICRV group, the estimated turning points cluster near 40% FSTS rather than migrating monotonically; the institutional regime governs the *strength of the concavity* more than the exact location of the peak. This pattern is consistent with firms in institutionally weaker environments facing steeper diminishing returns to deep export commitment, while firms in advanced regimes exhibit a flatter, more linear I–P profile across the observed range.

H5 is supported. The ICRV institutional regime systematically moderates the slope and curvature of the I–P function, with the concavity of the inverted-U deepening as institutional quality declines.

4.7 Full Three-Way Moderation (H6)

Table 2, Model M11.

The capstone model adds all three-way interactions (FSTS $\times$ DAI $\times$ ICRV, plus constituent lower-order interactions). The Lind–Mehlum test in M11 confirms the inverted-U (TP = 25.7%, LM p = .026), establishing that the inverted-U shape holds even in the most demanding specification. The DAI $\times$ ICRV interaction (β = +0.052, p = .049, *) is statistically significant, consistent with the CDCM prediction that digital capabilities provide stronger per-unit performance effects in weaker institutional environments. The three-way FSTS $\times$ DAI $\times$ ICRV term (β = −0.016, NS) is not significant, indicating that the DAI curve-reshaping effect documented in M8/M9 does not systematically vary across the ICRV regime gradient in this specification.

AdjR^2 = .057 in M11 (vs. .039 in M10) indicates meaningful incremental fit from adding the three-way moderation structure.

H6 (three-way interaction) partially supported; M11 inverted-U is confirmed. The DAI $\times$ ICRV main interaction is statistically significant (p = .049), supporting regime-contingent digital effects. The three-way FSTS curve-reshaping via DAI $\times$ ICRV does not reach statistical significance, the WBES DAI measure's limited dimensionality (Tier 1–2 only) likely constrains detection of the full CDCM-predicted interaction.

4.8 Multidimensional Performance Robustness

Firm performance is multidimensional (Combs, Crook, & Shook, 2005; Richard, Devinney, Yip, & Johnson, 2009), so we test whether the inverted-U survives an alternative operationalisation. Alongside the primary labour-productivity *level* outcome, we re-estimate the I–P relationship on real three-year **sales growth** (recovered from the WBES sales items, with country-year fixed effects absorbing within-wave inflation). The inverted-U is sharply identified for the productivity level (FSTS^2 β = −0.80, p = .008;

TP $\approx$ 37%) but is **not** detected for sales growth (FSTS^2 $\beta \approx 0$, p = .16; n.s.). This divergence is theoretically coherent rather than contradictory: the relocated-optimum mechanism concerns the *value captured per unit of internationalisation*, a stock property of productivity, not the flow dynamics of period-to-period growth. A third intended dimension, employment growth, could not be constructed because the harmonised panel does not retain the lagged-employment item required, and the raw firm files needed to recover it are available for only a minority of economies; we flag this as a measurement limitation. The primary inverted-U finding should therefore be read as a property of cross-sectional productivity levels, not of firm growth trajectories.

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5. Discussion

5.1 Theoretical Contributions: A Jointly-Determined Optimum

Our findings support an integrated account whose central claim is that the *form* of the I–P relationship is universal while its *optimum* is jointly determined by institutional and digital contingencies. This yields four contributions to international business theory.

First, the **relocated-optimum thesis** reframes the two-decade debate over the shape of the I–P curve. The inverted-U holds across all six ICRV regime groups (H1), but its shape is not a fixed structural parameter: the institutional regime governs the curve's *amplitude* (H5), a pronounced inverted-U in weaker-institution economies, flattening toward a near-linear profile in advanced ones, while digital capability reshapes the curve and relocates its effective optimum (H3). This dissolves the apparent contradiction between studies reporting positive-linear, inverted-U, and null effects (Section 5.2): they observe a single underlying curve whose sharpness, and whose payoff at high export intensity, are jointly set by institutional quality and digital capability. The contribution is to relocate the explanatory burden from the *existence* of non-linearity to the *determinants of its shape*, a shift from descriptive curve-fitting toward a contingency theory of where internationalization pays.

Second, **digital-for-institutional substitution** extends institutional-voids theory (Khanna & Palepu, 1997) into the digital domain. The DAI $\times$ ICRV interaction (β = +0.052, p = .049) shows that the curve-reshaping value of foundational digital adoption is greater precisely where formal institutions are weaker. Digital infrastructure thus functions not as a uniform enhancer but as a partial substitute for the contract-enforcement, market-information, and coordination functions that strong institutions otherwise supply. This is, to our knowledge, the first multi-economy firm-level evidence that digitalisation and institutional quality act as substitutes rather than complements in the internationalization-to-performance conversion.

Third, the **level–relocation distinction** separates two roles that the capabilities literature often conflates. Technological capability raises the performance level across the curve but does not reliably relocate the optimum in the pooled sample (Section 5.3), whereas institutional regime relocates the optimum without uniformly lifting the level. Distinguishing optimum-relocating forces from level-shifting forces clarifies why country-specific capability findings (e.g., TCI moderation in the companion Vietnam and China studies) need not generalise: capability-driven curve reshaping requires an institutional complementarity that is unevenly present across the ICRV spectrum.

Fourth, we offer the **ICRV framework** as a parsimonious contingency construct for cross-national IB research. By ordering economies jointly on institutional capability and resource vulnerability, dimensions that the varieties-of-capitalism (Hall & Soskice, 2001) and institutional-voids (Khanna & Palepu, 1997) traditions treat separately, ICRV predicts both the slope and the curvature of the I–P function across 45 economies, providing a transportable scaffold for theorising firm strategy under heterogeneous institutional regimes.

5.2 Resolution of Cross-Study Heterogeneity

The central finding, an inverted-U with a turning point at approximately 37% foreign sales intensity, robust across 45 economies and six ICRV groups, offers a parsimonious resolution to apparent inconsistencies in the Asian I–P literature. Studies finding positive linear effects (Pangarkar, 2008: Singapore; Cho et al., 2023: Korea) are working with samples centred well below the 37% turning point; their firms are on the ascending limb. Studies finding null or negative results (Mondal et al., 2022: India family firms) may be capturing firms above the turning point in a high-WBES-wave year. Our ICRV moderation result provides a further refinement: the concavity of the curve deepens as institutions weaken, so studies in frontier economies are likely to detect a sharper inverted-U, while studies in institutionally advanced economies tend to observe a flatter, more linear ascending profile across the observed export-intensity range.

5.3 Technology as Amplifier, Not Moderator

The finding that TCI raises the performance level without reliably reshaping the I–P curve in the pooled sample distinguishes the cross-economy result from country-specific evidence. In the companion Vietnam and China studies, TCI moderated the curve shape; in the 45-economy pool, institutional heterogeneity attenuates that moderation signal. This is consistent with the absorptive capacity argument (Cohen & Levinthal, 1990): high-TCI firms process international market signals more effectively, but the curve-shaping benefit requires institutional complementarity (e.g., strong IP protection, technology transfer infrastructure) that is inconsistently present across the full ICRV spectrum. The practical implication is that technological capability raises the performance floor universally, but its curve-reshaping role is context-contingent and cannot be assumed at scale.

5.4 Digital Adoption and Institutional Complementarity

The capability-complementarity curve reshaping found for DAI in M9, and the significant DAI $\times$ ICRV interaction (β = +0.052, p = .049) in M11, provides the first multi-economy evidence for the CDCM prediction that digital capabilities' performance effects are context-contingent. In institutionally stronger economies (Group I–II), digital infrastructure is near-universal and therefore provides competitive advantage only to early adopters at the margin; in weaker institutional environments (Group IV–VI), digital adoption remains differentiating. The implication is that policies promoting digitalisation in frontier economies are likely to generate stronger firm-level returns per unit of adoption than equivalent policies in advanced economies.

The pattern of significance across the two- and three-way terms is itself substantively informative and should be read as a structural boundary rather than a weakness of the evidence. The two-way DAI $\times$ ICRV interaction is significant (β = +0.052, p = .049) while the three-way FSTS $\times$ DAI $\times$ ICRV term is null: digital capability raises and protects the performance *level* more in weaker-institution regimes, but it does not detectably reshape the *curvature* of the I–P function across the institutional gradient. We interpret this asymmetry as the signature of substitution operating at the level rather than the shape. The substitution mechanism, foundational digital tools standing in for the contract-enforcement and market-information functions that strong institutions otherwise supply, is powerful enough to lift the productivity floor where institutions are thin, but the surface-level Tier-1 and Tier-2 capabilities the WBES captures (a static website, an electronic-payment facility) are too thin an instrument to bend the second-order curvature of the export-performance trade-off in high-friction markets. The two-way result is therefore best read as a strong directional signal at the level margin rather than a settled magnitude, and the null three-way as a boundary on what surface digitalisation can accomplish, not as a refutation of the substitution logic. Reading the borderline p = .049 with appropriate caution, we frame the substitution claim as the most theoretically consequential and empirically supported component of the digital story, while leaving the curvature-reshaping potential of richer (Tier 3–4) digital capabilities as an open question for future data.

5.5 Managerial Heterogeneity

The consistent and significant performance premium for female top managers (approximately 22–23% across specifications) is notable given the diversity of economies in our sample, from conservative Gulf economies (Bahrain, Iraq) to progressive innovation hubs (Korea, Singapore, Israel). This cross-context robustness supports the view that female top management is a genuine resource (Hambrick & Mason, 1984; Richard et al., 2019), not merely a proxy for firm governance quality. The null curve-moderation result (FSTS$\times$female_manager not significant) suggests that the performance premium is universal across FSTS levels, female managers improve absolute performance but do not help firms navigate internationalization more efficiently.

Two recent World Bank studies corroborate the premium from a supply-side perspective. In Vietnam, one of our largest ICRV Group IV contributors, with a female-to-male labour force participation ratio of 88.61% (World Bank Prosperity Data360, 2025), the *Care for Growth* policy note (Buchhave et al., 2026) documents that women who hold management positions have cleared exceptional retention barriers: childbirth reduces women's wage employment probability by 8.1 percentage points and urban household income by 27%, with no significant effect on fathers. Women who reach managerial rank therefore represent a highly selected and committed workforce stratum, consistent with the productivity premium we estimate. At the sector level, IFC (2022) similarly documents that female leadership in Vietnamese banking is associated with superior risk management and stakeholder orientation, the relational-capital mechanisms posited by Upper Echelons Theory (Hambrick & Mason, 1984). These convergent pieces of evidence strengthen the interpretation that the WBES-estimated premium reflects genuine managerial capability rather than endogenous selection into high-productivity firms.

5.6 Limitations

Five inferential constraints bound these findings. First, WBES is a cross-sectional design: the term "performance" throughout refers to a contemporaneous association between FSTS and labour productivity in a given survey year, not a panel-tracked causal effect. The IV strategy used in the companion Vietnam study to address selection into exporting is not available at scale across all forty-five economies, and as Section 2.4 notes the negative ascending-limb interaction for digital adoption cannot be fully separated from a selection-driven reading; the level effects, against which selection works, are the more securely interpretable estimates.

Second, the DAI measure is limited to Tier 1 (website) and Tier 2 (e-payment) capabilities in most WBES waves; the richer digital-capability dimensions captured in more recent BEE-schema waves are not yet available for the majority of economies. This measurement floor is the most plausible explanation for the null three-way curvature interaction (Section 5.4) and bounds the substitution claim to the surface-digital layer.

Third, the ICRV construct, although central to the contribution, is an author-constructed ordinal scale and warrants validation scrutiny. The strongly positive ICRV level coefficient (β = +0.592) reflects higher nominal sales-per-worker in lower-price-level economies rather than a substantive institutional level effect, so ICRV partly co-varies with the development and price level of an economy. We mitigate this in two ways, by absorbing price levels through country-year fixed effects in the level benchmark (M5) and by reading the institutional claim off the *interaction* terms rather than the main effect; nevertheless, the ordinal scale cannot fully disentangle institutional quality from broader development. Future work should validate ICRV against external governance indices (for example, the Worldwide Governance Indicators) and test whether a continuous institutional-quality measure reproduces the concavity gradient.

Fourth, the sampling frame is deliberately Asia-Pacific, so the relocated-optimum and substitution results should be generalised to other regions only with caution until replicated; the transcontinental-exclusion robustness check (Section 3.1) guards internal consistency but not external transportability. Fifth, productivity and the focal predictors are drawn from the same firm survey instrument, raising a common-source concern, although the objective, register-style nature of the items (sales, worker counts, certification, website presence) limits its severity.

A final coverage note: the dataset covers 45 ICRV-mapped economies across six regime groups; Oman's only available wave (2003) lacks the core productivity and export-intensity items and so does not enter the estimation sample. Future waves of the Oman WBES would sharpen Group II estimates.

5.7 Practical and Policy Implications

The relocated-optimum logic translates into a sequencing rule for managers rather than a uniform prescription to internationalise. Because the performance-maximising export intensity sits near 37% of sales in the pooled frame, firms below that threshold are on the ascending limb, where the binding task is removing barriers to export expansion and the marginal return to additional foreign sales is positive. Firms already beyond their regime-specific optimum face a different problem: the constraint is coordination capacity, not market access, and further export commitment erodes productivity, most steeply in weaker-institution regimes where the concavity is sharpest. The same model therefore prescribes opposite actions, expand versus consolidate, depending on a firm's position on a curve whose location managers can in principle estimate from their own export share and institutional context.

For digital strategy, the substitution result implies that foundational digitalisation is a higher-leverage instrument precisely where institutions are weakest. In frontier and vulnerable regimes, basic digital interfaces partly stand in for the contract-enforcement and market-information functions that thin institutions fail to provide, so the per-unit performance return to adoption is largest there; in digitally saturated advanced economies the same investment is closer to a competitive-parity hygiene factor. Managers in weak-institution settings should thus sequence foundational digital adoption alongside, not after, export expansion, while those in advanced economies should look beyond Tier 1–2 tools to deeper digital capabilities for differentiation. Technological capability, by contrast, is best read as a baseline performance lever that lifts the floor across all export intensities rather than a curve-shifting intervention.

For policy, the findings sharpen where digitalisation support yields the highest social return. Programmes that subsidise or de-risk foundational firm digital adoption are likely to generate stronger firm-level productivity gains in frontier and small-island economies than equivalent programmes in advanced ones, making digital public infrastructure a complement to, and partial near-term substitute for, the slower work of institutional reform. At the same time, the steep descending-limb penalty in weak-institution regimes cautions against export-promotion policies that push firms toward maximal foreign-sales intensity irrespective of coordination capacity; institutional development that flattens the concavity remains the first-best lever for raising the returns to deep internationalisation.

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6. Conclusions

This study set out to explain not whether internationalization pays, but where and for whom. Across a 45-economy, 98-wave WBES sample, the inverted-U I–P relationship is universal in form, peaking at approximately 37% export intensity, yet its optimum is jointly conditioned by institutional regime and digital capability, institutions governing the curve's amplitude and digital capability reshaping it, with digital capability operating as a partial substitute for institutional strength. Three implications follow. First, for firms on the ascending limb (FSTS < 37%), intensifying internationalization is productivity-enhancing and the priority is removing barriers to export expansion; beyond the optimum, the binding constraint is coordination capacity rather than market access. Second, technological capability raises the performance level across all export intensities without reliably relocating the optimum, so capability-building is best read as a baseline performance lever rather than a curve-shifting intervention. Third, foundational digital adoption is both a level enhancer and a curve reshaper, and its returns are largest where institutions are weakest (DAI $\times$ ICRV, $p = .049$), making digitalisation a particularly high-leverage instrument in frontier and vulnerable regimes.

The relocated-optimum logic also clarifies which firms are most exposed to macroeconomic shocks: those pushed above their regime-specific optimum by earlier export commitments face the steepest descending-limb penalties, which the ICRV results show are sharpest in weaker-institution regimes. Recent macro-monitoring for Vietnam, the sample's largest Group IV economy, illustrates the point, with contracting export orders in early 2026 (World Bank, 2026b) placing high-FSTS exporters on exactly this exposed segment. Digital capability can buffer the descent by flattening the descending limb (M8–M9), providing a firm-level adjustment margin when external demand weakens.

Future research should leverage the growing panel dimensions of WBES to establish causality, incorporate Tier 3–4 digital capability measures (AI adoption, cloud computing, platform participation) as these items become available in the 2025+ WBES waves, and test directly whether the digital-for-institutional substitution documented here strengthens or decays as institutions develop.

Data Availability and Acknowledgements

The microdata analysed in this study are publicly available from the World Bank Enterprise Surveys (www.enterprisesurveys.org). The authors gratefully acknowledge the World Bank Enterprise Analysis Unit for collecting and curating the data; the findings, interpretations, and conclusions are those of the authors and do not necessarily reflect the views of the World Bank. Harmonisation code and replication materials are available from the authors on reasonable request.

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Appendix A: Model Sequence

Table 1: Variable definitions and measurement

Variable	Definition	Source items	Coverage
ln(LP)	Log labour productivity (sales / workers)	n3 or d2; l1	~85%
FSTS	Foreign sales share (direct exports)	d3c / 100	~83%
TCI_z	Tech capability index (cert + foreign tech), z-scored	b8, e6	~82%
DAI_z	Digital adoption (website + e-pay share), z-scored	c22b/e1, k33	~83%
mgr_experience	Top manager years in sector	b7	~84%
mgr_female	Top manager is female (binary)	b7a, b6a	~82%
female_owner	Majority female ownership (binary)	b4	~99%
foreign_own_pct	Foreign ownership %	b6a / 100	~52%
firm_age	Years since establishment	current yr − b5	~64%
ln_size	Log permanent workers	l1	~84%
ICRV_group	Institutional regime group (1–6)	ICRV classification	100%

Table 2: Model fit and turning points

Model	N	Adj R^2	Turning point (%)	LM p	Key additions
M0	82,302	.000	—	—	FSTS linear
M1	82,302	.000	—	—	FSTS + controls (no quadratic)
M2	82,302	.002	**37.1**	<.001	FSTS + $FSTS^2$

Model	N	Adj R^2	Turning point (%)	LM p	Key additions
M3	36,772	.017	**34.1**	<.001	M2 + controls
M5	36,772	.671	**39.5**	<.001	M3 + country-year FE
M6	36,481	.027	33.0	<.001	M3 + TCI (main only)
M7	36,481	.028	**34.2**	<.001	M6 + FSTS×TCI, FSTS²×TCI
M8	36,372	.033	**34.2**	<.001	M7 + DAI + FSTS×DAI, FSTS²×DAI
M9	34,228	.042	**32.2**	<.001	M8 + manager + interactions
M10	30,360	.039	—	n.s.	M8 + ICRV + FSTS×ICRV, FSTS²×ICRV
M11	28,500	.057	**25.7**	.026	M10 + three-way DAI×ICRV×FSTS

Note. Turning point confirmed by Lind–Mehlum (2010) test (LM p). M10 turning point not identified because ICRV interactions shift the inflection point outside the sample range for some groups. HC1 robust SEs throughout.

Table 3: Key coefficient estimates (M2, M7, M8, M10)

Term	M2	M7	M8	M9	M10
FSTS (β_1)	+1.211***	+2.249***	+2.154***	+2.429***	−6.497***
FSTS² (β_2)	−1.630***	−3.287***	−3.147***	−3.768***	+10.449***
TCI_z	—	+0.372***	+0.330***	+0.326***	+0.279***
FSTS×TCI	—	−0.090	+0.159	+0.158	—
FSTS²×TCI	—	−0.227	−0.493†	−0.433	—

Term	M2	M7	M8	M9	M10
DAI_z	—	—	+0.172***	+0.160***	+0.198***
FSTS×DAI	—	—	−0.588***	−0.786***	—
FSTS²×DAI	—	—	+0.726**	+0.982***	—
mgr_experience	—	—	—	+0.018***	—
mgr_female	—	—	—	+0.202***	—
ICRV_group	—	—	—	—	+0.592***
FSTS×ICRV	—	—	—	—	+1.849***
FSTS²×ICRV	—	—	—	—	−2.932***
N	82,302	36,481	36,372	34,228	30,360
Adj R²	.002	.028	.033	.042	.039

Note. HC1 robust SEs. *** p < .001, ** p < .01, * p < .05, † p < .10. Controls (female_owner, foreign_own_pct, firm_age, ln_size) included in M7–M10 but not reported for space. M11 (N=28,500, AdjR^2=.057, TP=25.7%, LM p=.026): DAI × ICRV=+0.052, *three-way FSTS × DAI × ICRV=−0.016 (NS), mgr_experience=+0.022*, **FSTS×mgr=+0.007 (NS), mgr_female=+0.210***, female_owner=+0.122**.

Appendix B: TFP Production Function Estimates (Robustness Reference)

Sector-level production function coefficients used to construct alternative TFP-based dependent variables for robustness checks. Two specifications are available: VA=f(K,L) (VAKL, value-added formulation) and Y=f(K,L,M) (YKLM, gross-output formulation), each estimated separately for 20 two-digit ISIC manufacturing sectors with high-income economy interaction terms and year fixed effects (2009–2025). Coefficients and t-statistics are stored in `replication/Annex_TFP_regression_tables_March_11_2026.xlsx`. TFP residuals from these regressions can be used as an alternative dependent variable to ln(LP) to test whether the inverted-U relationship is robust to output-per-worker vs. total-factor productivity operationalisations.

[Author identifying information removed for blind review.]